

Quantum Systems

(Lecture 3 - conclusion)

(The principles of quantum computation II: Measurement)

Luís Soares Barbosa



Universidade do Minho



Universidade do Minho

The principles

Quantum computation explores the laws of quantum theory as computational resources.

Thus, the principles of the former are directly derived from the postulates of the latter.

- The state representation postulate
- The state evolution postulate
- The state composition postulate
- The state measurement postulate

The measurement postulate

Postulate 4

For a given orthonormal basis $B = \{|v_1\rangle, |v_2\rangle, \dots\}$, a measurement of a state space $|\nu\rangle = \sum_i \alpha_i |v_i\rangle$ wrt B , outputs the label v_i with probability $\|\alpha_i\|^2$ and leaves the system in state $|v_i\rangle$.

- Given a state

$$|\nu\rangle = \sum_i \alpha_i |v_i\rangle$$

the probability of collapsing to base state $|v_i\rangle$ is $\|\langle v_i | \nu \rangle\|^2$.

- Measurements are made through **projectors** which identify the 'data' (i.e. the subspace of the relevant Hilbert space where the quantum system lives) one wants to measure.

Projectors

Any **projector** P identifies in the state space V a subspace V_P of all vectors $|\phi\rangle$ that are left unchanged by P , i.e. such that

$$P|\phi\rangle = |\phi\rangle$$

Examples

- The **identity** I projects onto the whole space V .
- The **zero operator** projects onto the space $\{0\}$ consisting only of the zero vector.
- $|\nu\rangle\langle\nu|$ is the projector onto the subspace spanned by $|\nu\rangle$.

Outer product

- **inner product** $\langle w|v\rangle$: multiplying $|v\rangle$ on the left by the dual $\langle w|$, yields a scalar.
- **outer product** $|w\rangle\langle v|$: multiplies on the right, yielding an operator:

$$|w\rangle\langle v| (|u\rangle) = |w\rangle\langle v|u\rangle = \langle v|u\rangle |w\rangle$$

Clearly

$$|v\rangle\langle v| (|u\rangle) = \langle v|u\rangle |v\rangle$$

which projects $|u\rangle$ to the 1-dimensional subspace of H spanned by $|v\rangle$

Projectors

Examples

- Projector $|0\rangle\langle 0|$ projects onto the subspace generated by $|0\rangle$, i.e.

$$|0\rangle\langle 0|(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle\langle 0|(|0\rangle) + \beta|0\rangle\langle 0|(|1\rangle) = \alpha|0\rangle$$

- Similarly, $|10\rangle\langle 10|$ acts on a two-qubit state

$$v = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

yielding

$$|10\rangle\langle 10|(|v\rangle) = \alpha_{10}|10\rangle$$

and

$$|00\rangle\langle 00| + |10\rangle\langle 10|(|v\rangle) = \alpha_{00}|00\rangle + \alpha_{10}|10\rangle$$

Mathematically, what is a projector?

?

- ... an **operator** (of course, whatever could it be?)
- but not a **unitary** one ...

Projectors

A projector $P : V \rightarrow V_P$ is an operator such that

$$P^2 = P$$

Additionally, we require P to be Hermitian, i.e.

$$P = P^\dagger$$

Note that the combination of both properties yields

$$\|P|v\rangle\|^2 = (\langle v|P^\dagger)(P|v\rangle) = \langle v|P|v\rangle$$

Example

The probability of getting state $|0\rangle$ when measuring $\alpha|0\rangle + \beta|1\rangle$ with $P = |0\rangle\langle 0|$ is computed as

$$\|P|v\rangle\|^2 = \langle v|P|v\rangle = \langle v|0\rangle\langle 0|v\rangle = \langle v|0\rangle\langle 0|v\rangle = \bar{\alpha}\alpha = \|\alpha\|^2$$

Projectors

Two projectors P, Q are **orthogonal** if $PQ = 0$.

The sum of any collection of **orthogonal** projectors $\{P_1, P_2, \dots\}$ is still a projector (verify!).

A projector P has a **decomposition** if it can be written as a sum of **orthogonal** projectors:

$$P = \sum_i P_i$$

Such projectors yield **measurements** wrt to the corresponding decomposition.

Examples

- **Complete** measurement in the computational basis wrt to decomposition

$$I = \sum_{i \in 2^n} |i\rangle\langle i|$$

in a state with n qubits.

- **Incomplete** measurement: e.g.

$$\sum_{\{i \in 2^n \mid i \text{ even}\}} |i\rangle\langle i|$$

Projectors

Example: measuring up to (bit equality)

$$V = S_e \oplus S_n$$

with S_e the subspace generated by $\{|00\rangle, |11\rangle\}$ in which the two bits are equal, and S_n its complement. P_e and P_n , are the corresponding projectors.

When measuring

$$v = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

with this device, yields a state in which the two bit values are equal with probability

$$\langle v | P_e | v \rangle = (\sqrt{\|\alpha_{00}\|^2 + \|\alpha_{11}\|^2})^2 = \|\alpha_{00}\|^2 + \|\alpha_{11}\|^2$$

Of course, the measurement does not determine the value of the two bits, only whether the two bits are equal

Projectors

Any orthonormal collection of vectors $B = \{|v_1\rangle, |v_2\rangle, \dots\}$ defines a projector

$$P = \sum_i |v_i\rangle\langle v_i|$$

If B spans the entire Hilbert space V , it forms a **basis** for V and $P = I$, i.e. B provides a **decomposition** for the identity.

Projectors

Is there a standard way to provide a decomposition for P ?

Yes, if P is a **Hermitian** operator, because of the

Spectral theorem

Any Hermitian operator on a finite Hilbert space V provides a basis for V consisting of its **eigenvectors**.

Projectors are Hermitian

Hermitian operators

- define a unique orthogonal subspace decomposition, their **eigenspace decomposition**, and
- for every such decomposition, there exists a corresponding Hermitian operator whose eigenspace decomposition coincides with it

Projectors are Hermitian

Property 1

Every eigenvalue λ with eigenvector $|r\rangle$ is **real**, because

$$\lambda \langle r|r \rangle = \langle r|\lambda|r \rangle = \langle r|(P|r) \rangle = (\langle r|P^\dagger)|r \rangle = \bar{\lambda} \langle r|r \rangle$$

Property 2

The eigenvectors for distinct eigenvalues must be **orthogonal**, because

$$\lambda \langle v|w \rangle = (\langle v|P^\dagger)|w \rangle = \langle v|(P|w) \rangle = \mu \langle v|w \rangle$$

for any pairs $(\lambda, |v\rangle), (\mu, |w\rangle)$ with $\lambda \neq \mu$.

Thus, $\langle v|w \rangle = 0$, because $\lambda \neq \mu$, and the corresponding subspaces are orthogonal.

Projectors are Hermitian

Eigenspace decomposition of V for P

Any Hermitian P determines a unique decomposition for V

$$V = \oplus_{\lambda_i} S_{\lambda_i}$$

and any decomposition $V = \oplus_{i=1}^k S_i$ can be realized as the eigenspace decomposition of a Hermitian operator

$$P = \sum_i \lambda_i P_i$$

where each P_i is the projector onto S_{λ_i}

Projectors are Hermitian

A decomposition can be specified by a Hermitian operator

- Any measurement is specified by a Hermitian operator P
- The possible outcomes of measuring a state $|v\rangle$ with P are labeled by the eigenvalues of P
- The probability of obtaining the outcome labelled by λ_i is

$$\|P_i|v\rangle\|^2$$

- The state after measurement is the normalized projection

$$\frac{P_i|v\rangle}{\|P_i|v\rangle\|}$$

onto the λ_i -eigenspace S_i . Thus, the state after measurement is a unit length eigenvector of P with eigenvalue λ_i

Projectors are Hermitian

Notes

- A measurement is not modelled by the action of a Hermitian operator on a state, but of the corresponding projectors.
- Actually, Hermitian operators are only a bookkeeping trick
- A Hermitian operator uniquely specifies a subspace decomposition
- For a given subspace decomposition there are many Hermitian operators whose eigenspace decomposition is that decomposition.

Projectors are Hermitian

Example: Measuring a single qubit in the Hadamard basis

Operator

$$X = |0\rangle\langle 1| + |1\rangle\langle 0| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is Hermitian, with eigenvalues $\lambda_+ = 1$ and $\lambda_- = -1$, and $|+\rangle, |-\rangle$ the corresponding eigenvectors, thus yielding the following projectors:

$$P_+ = |+\rangle\langle +| = \frac{1}{2}(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 1|)$$

$$P_- = |-\rangle\langle -| = \frac{1}{2}(|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|)$$